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WEI Shiyi

Research Interests

My research interests are in nonequilibrium statistical physics and disordered systems, quantum diffusion and mesoscopic transport, density and current fluctuations in noisy quantum systems and large deviations and macroscopic fluctuation theory.

Education

Université Côte d'Azur — Nice, France
Ph.D. in Theoretical Physics — Nov. 2024 – Oct. 2027 (expected)
Thesis director: Mathias Albert; Co-supervisor: Tony Jin

- Initial thesis project: Towards a typical theory of quantum diffusion.

École Normale Supérieure — Paris, France
Diplôme de l'ENS, Physics — 2020 – 2024

- Master 1 and Master 2 ICFP, Quantum Physics track.

Sorbonne Université — Paris, France
Licence 1 and 2, Mathematics and Physics (double major) — 2018 – 2020

Publications

1. Mathias Albert, Denis Bernard, Tony Jin, Stefano Scopa and **Shiyi Wei**. "Universal classical and quantum fluctuations in the large deviations of current of noisy quantum systems: Quantum symmetric simple exclusion and inclusion processes." *Physical Review E* **114**, 034140 (2026). doi:10.1103/dt4c-tz98; arXiv:2601.16883.

2. Denis Bernard, Tony Jin, Stefano Scopa and **Shiyi Wei**. "Large deviations of density fluctuations in the boundary-driven quantum symmetric simple inclusion process." *Physical Review E* **112**, 034106 (2025). doi:10.1103/fs7f-z3k6; arXiv:2503.18763.

Research Experience

Institut de Physique de Nice (INPHYNI) — Nice, France
Doctoral researcher — Nov. 2024 – present

Advisors: Mathias Albert and Tony Jin

- Theoretical study and analytical and numerical modelling of fluctuations in nonequilibrium disordered systems.

Institut de Physique de Nice (INPHYNI) — Nice, France
Master 2 research intern — Apr. – Jun. 2024

Advisors: Mathias Albert and Tony Jin

- Analytical calculation of nonequilibrium current cumulants in the quantum symmetric simple exclusion process (QSSEP), using the Lindblad formalism and quantum Itô calculus.

LPMMC (CNRS) — Grenoble, France
Master 1 research intern — Feb. – Jul. 2022

Advisor: Anna Minguzzi

- Modelled a Tonks–Girardeau gas and numerically simulated its density matrix in Mathematica.

Laboratoire Kastler Brossel (LKB) — Paris, France
Undergraduate research intern — Jun. – Jul. 2019

Advisor: Alice Sinatra

- Studied quantum-state reconstruction using Wigner distributions, coherent states and quantum squeezing.

Languages

- **French:** Fluent
- **English:** Fluent
- **Chinese:** Native